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# Project 3

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## Part A.1

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I took photos of the view of my desk, and at two different times with the idea to merge the two timepoints (which may be done in 3B)



ORG



WARP





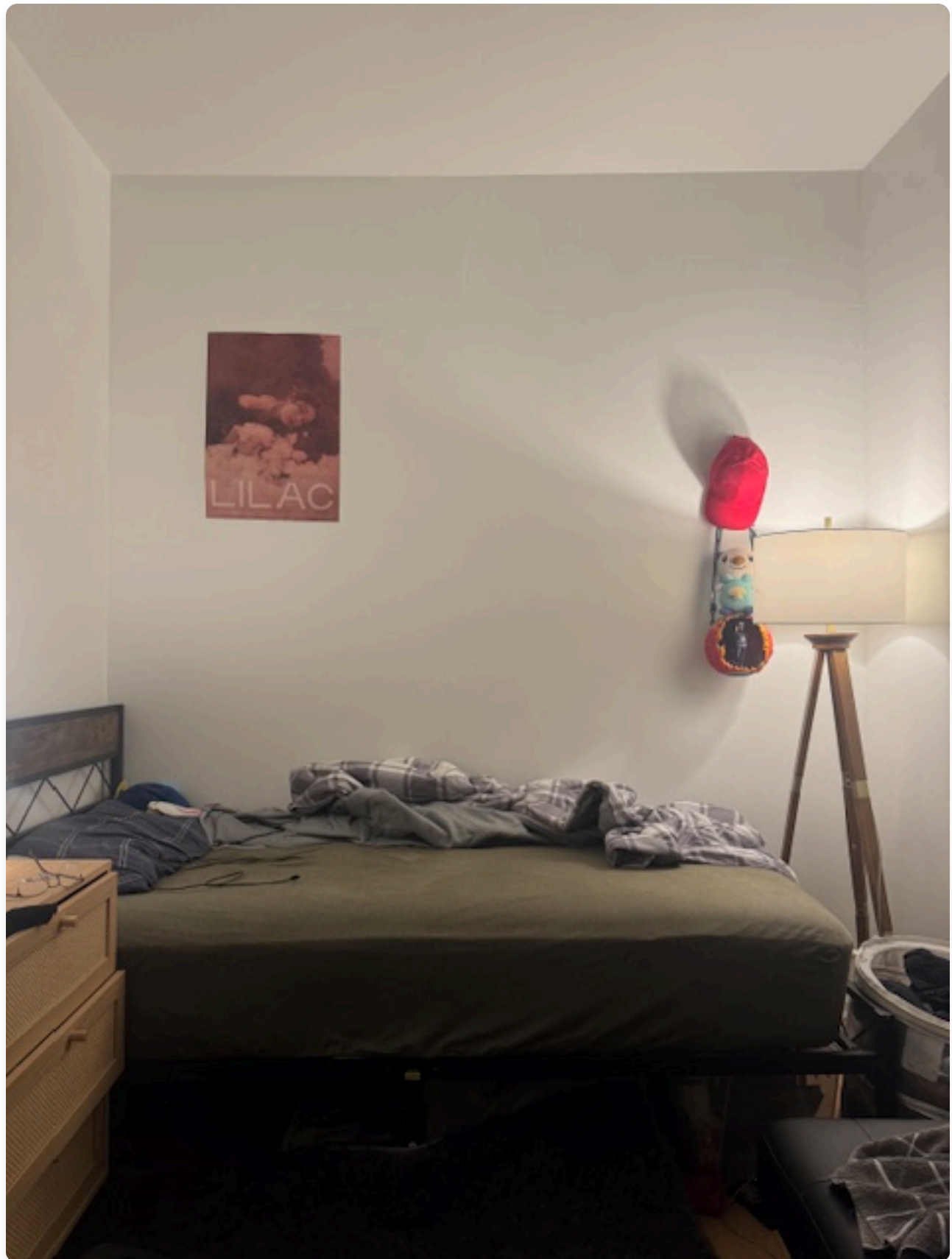
ORG



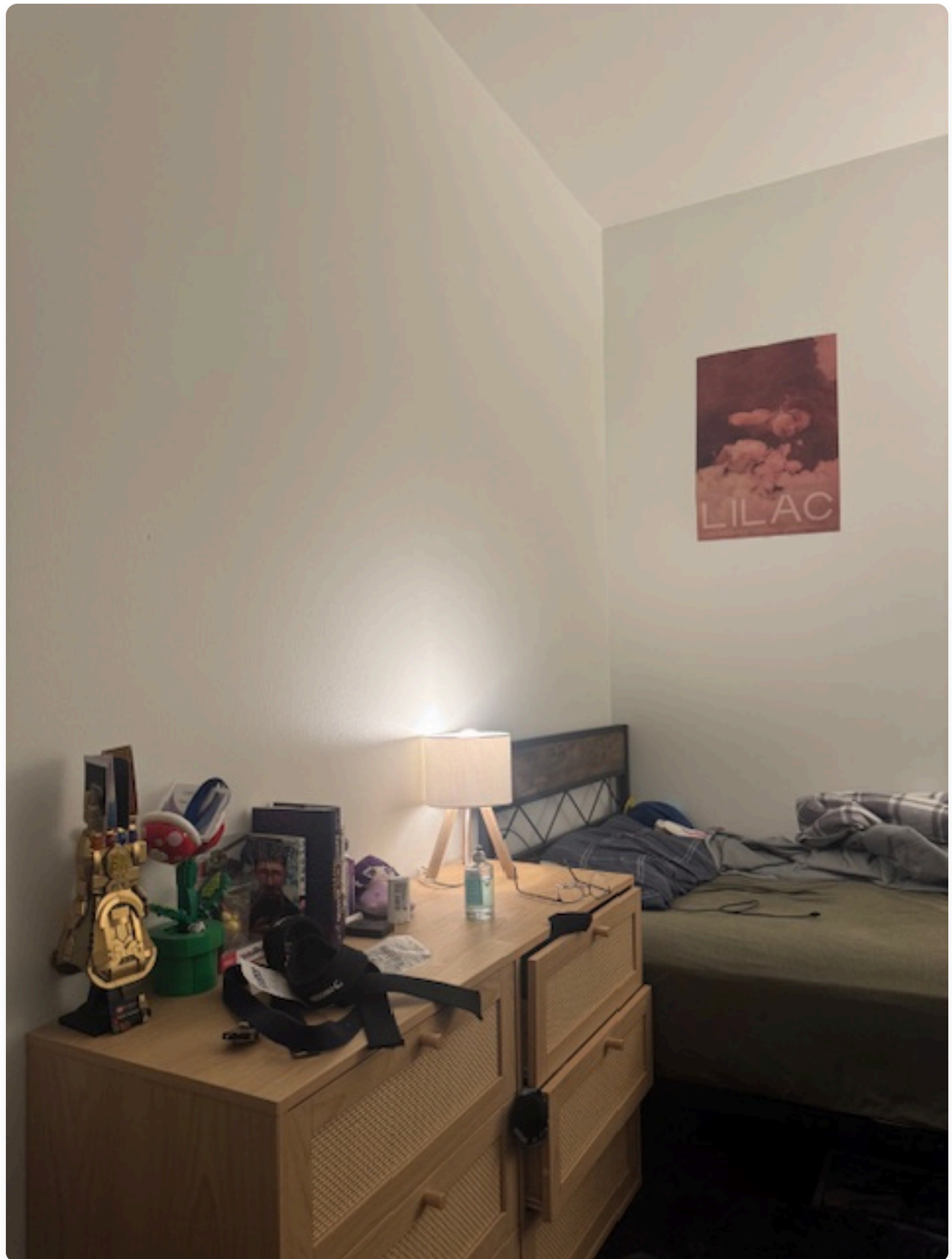


WARP

I also took a picture of my room to make a mosaic.



ORG



WARP



## Part A.2 Recover Homographies

I solved for the homographies in discussion and I'll show the results here.

homographie... 1 / 1 38%

**CS180/280A: Intro to Comp. Vision and Comp. Photography Discussion #5**

(a) We can start by algebraically eliminating the  $\lambda$  term. Write expressions for  $u$  and  $v$  in terms of  $h_1, \dots, h_8$ , and  $x, y$ . (hint: division)

$$\lambda u = h_1 x + h_2 y + h_3$$

$$\lambda v = h_4 x + h_5 y + h_6$$

$$\lambda = h_7 x + h_8 y + 1$$

$$u = \frac{h_1 x + h_2 y + h_3}{h_7 x + h_8 y + 1}$$

$$v = \frac{h_4 x + h_5 y + h_6}{h_7 x + h_8 y + 1}$$

(b) Convert your two equations from Part (a) into linear equations in the form:  $(\dots)h_1 + (\dots)h_2 + \dots + (\dots)h_8 = (\dots)$ . Why is this useful?

$$h_1 x + h_2 y + h_3 - h_7 x u - h_8 y u = u$$

$$h_4 x + h_5 y + h_6 - h_7 x v - h_8 y v = v$$

(c) What  $a_1^T, a_2^T, b_1$ , and  $b_2$  values does this correspondence pair produce?

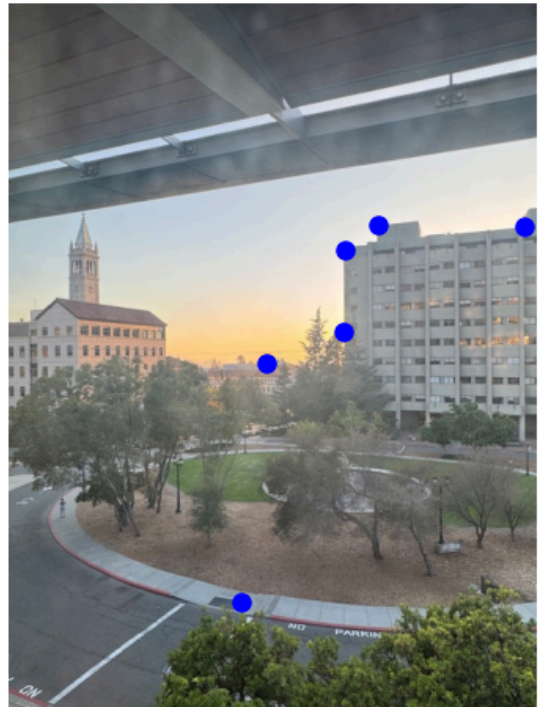
Homographies calculation (PDF)

Now here are the correspondents I chose for Evans, and the recovered matrix.

Image 0



Image 2



Evans: Chosen Correspondences

```
[ 5.71100148e-01,  3.13770620e-02,  1.87975122e+02]  
[-2.74463231e-01,  9.12438766e-01,  3.52833705e+01]  
[-8.60347724e-04,  7.81272190e-05,  1.00000000e+00]
```

Recovered Homography Matrix

## Part A.3 Warp images

I took a picture of my lab to warp and make my photo square, and also found this minecraft patinting online which I hope to warp so that the image is facing me

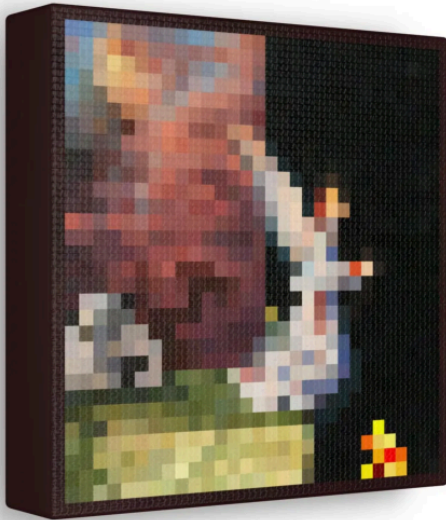
Lab Photo (Original)



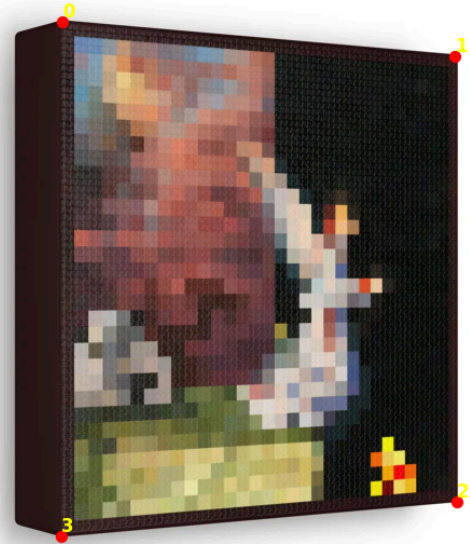
Lab Photo with Corner Points Labeled



Minecraft Painting (Original)



Minecraft Painting with Corner Points Labeled



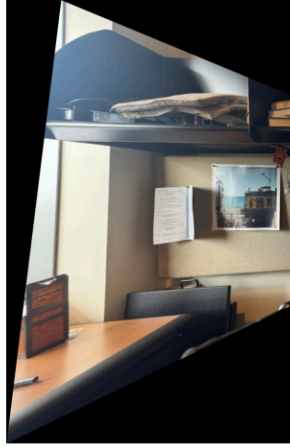
Original image to warp



Original Lab Photo



Warped Lab Photo (Bilinear)

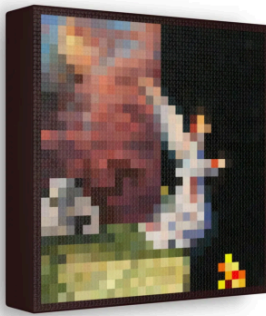


Warped Lab Photo (Nearest Neighbor)

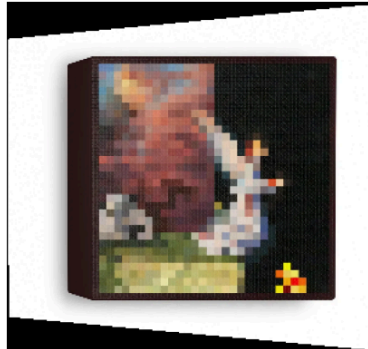


Warped lab image (made square)

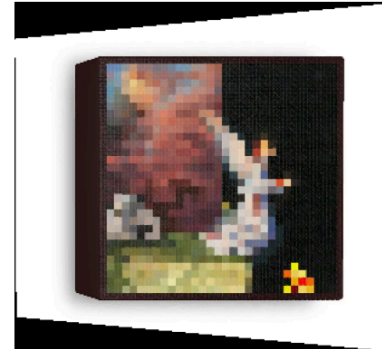
Original Minecraft Painting



Warped (Bilinear)



Warped (Nearest Neighbor)



Warped Minecraft painting

## Part A.4

I warp the corners of the image to fit, then use that to calculate the offset to build the mosaic, then I use this aligned image and apply weighted mean blending.

p>





Evans sunset org



Evans sunset warp



Evans Mosaic 1

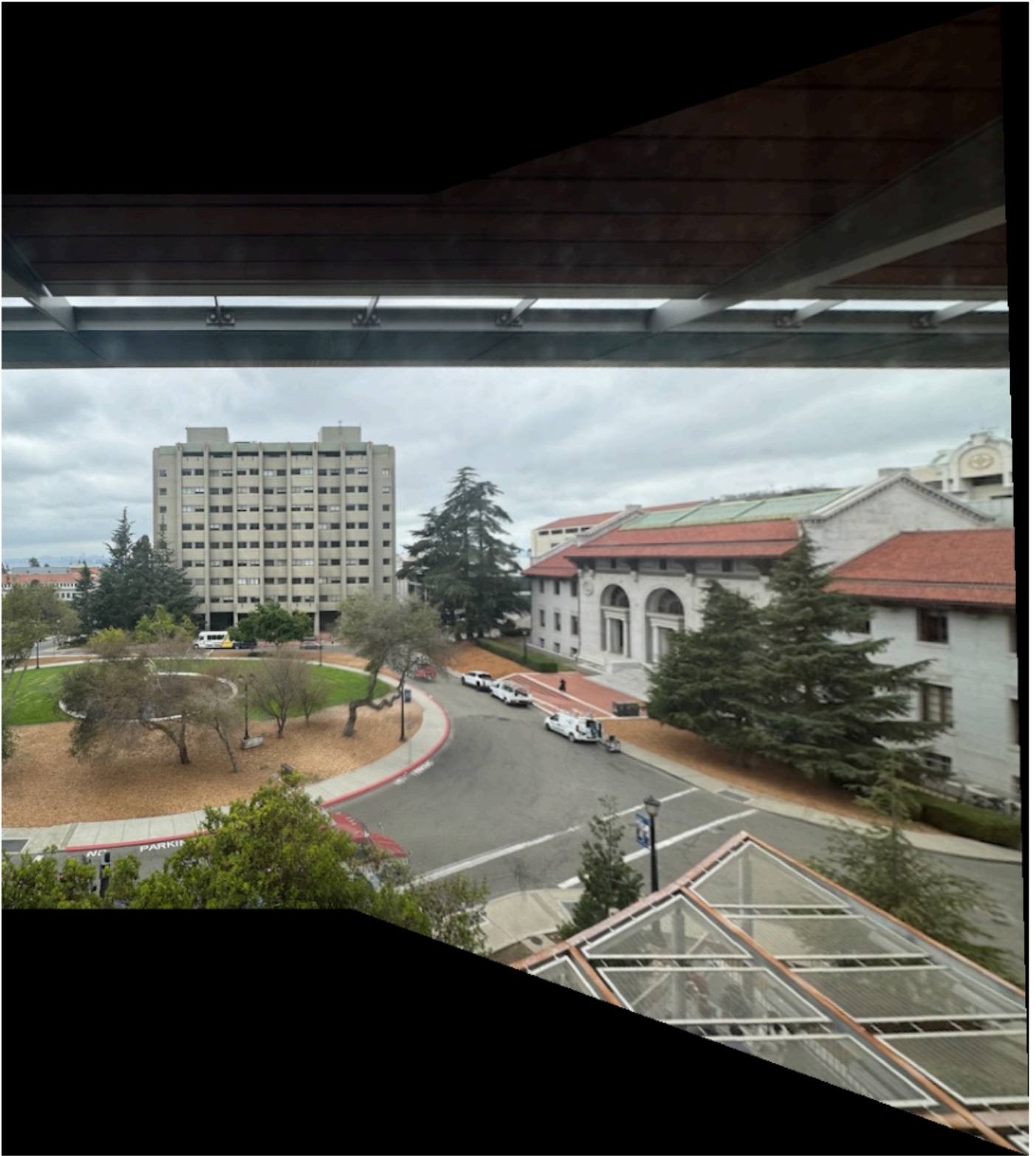




Evans foggy org

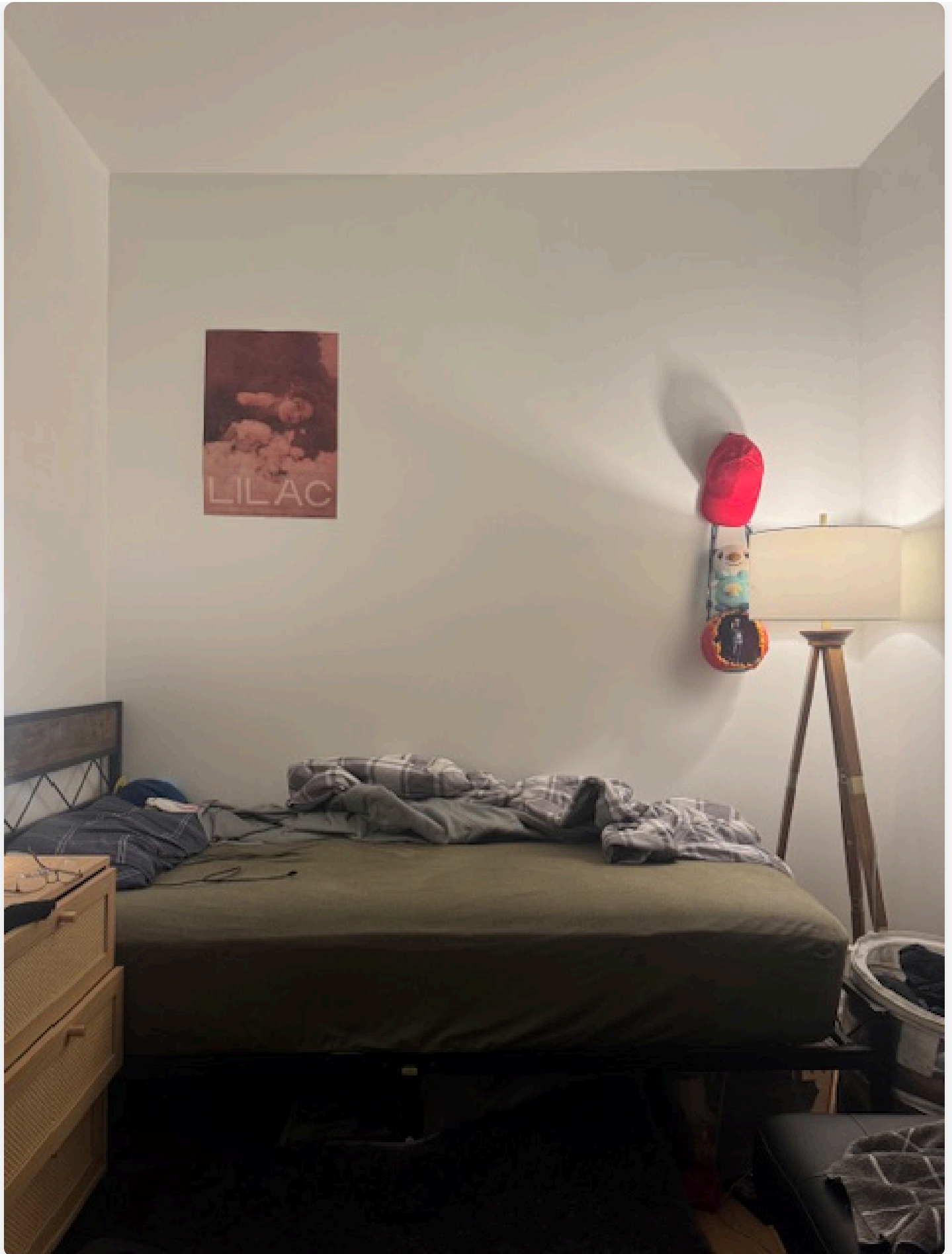


Evans foggy warp

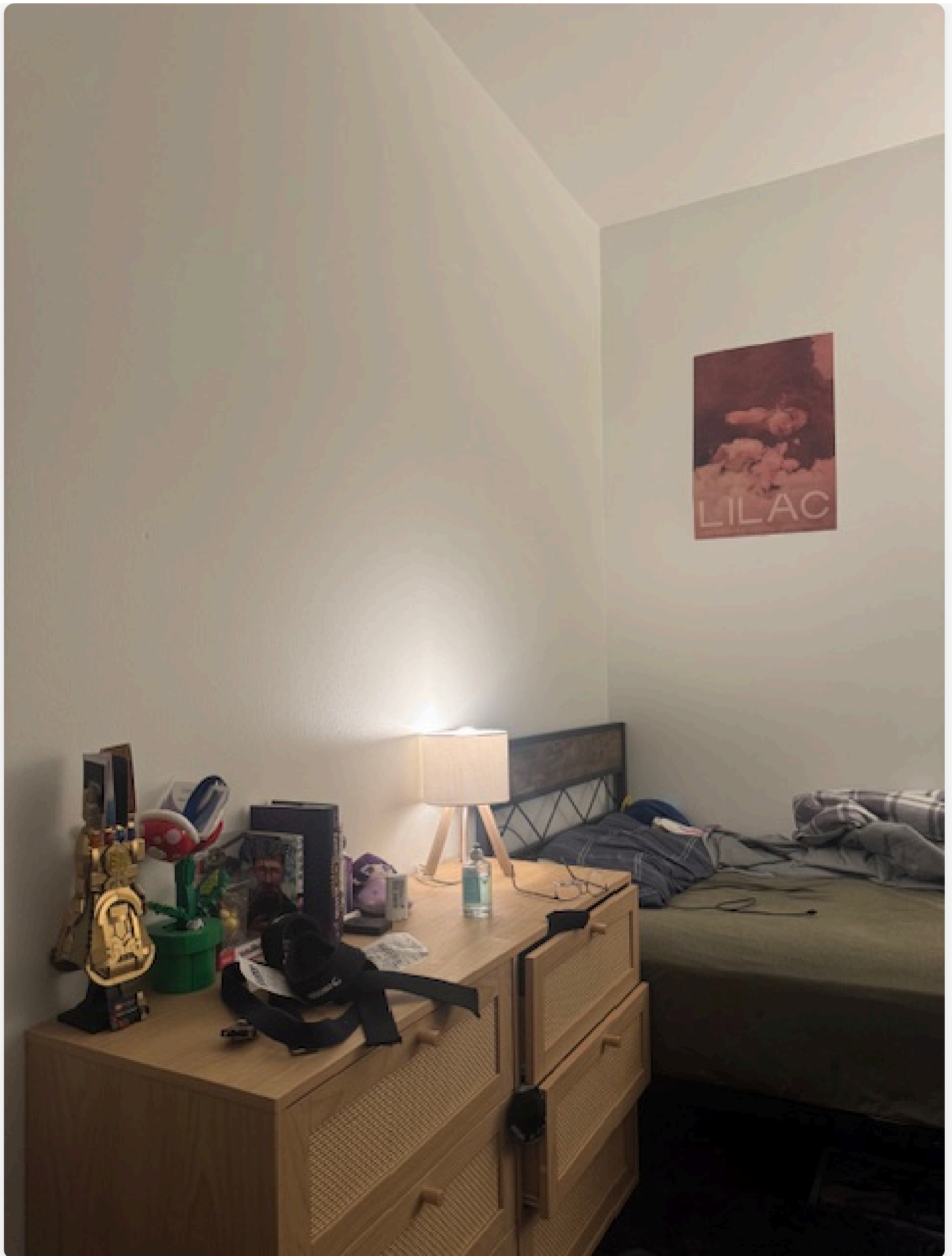


Evans Mosaic 2

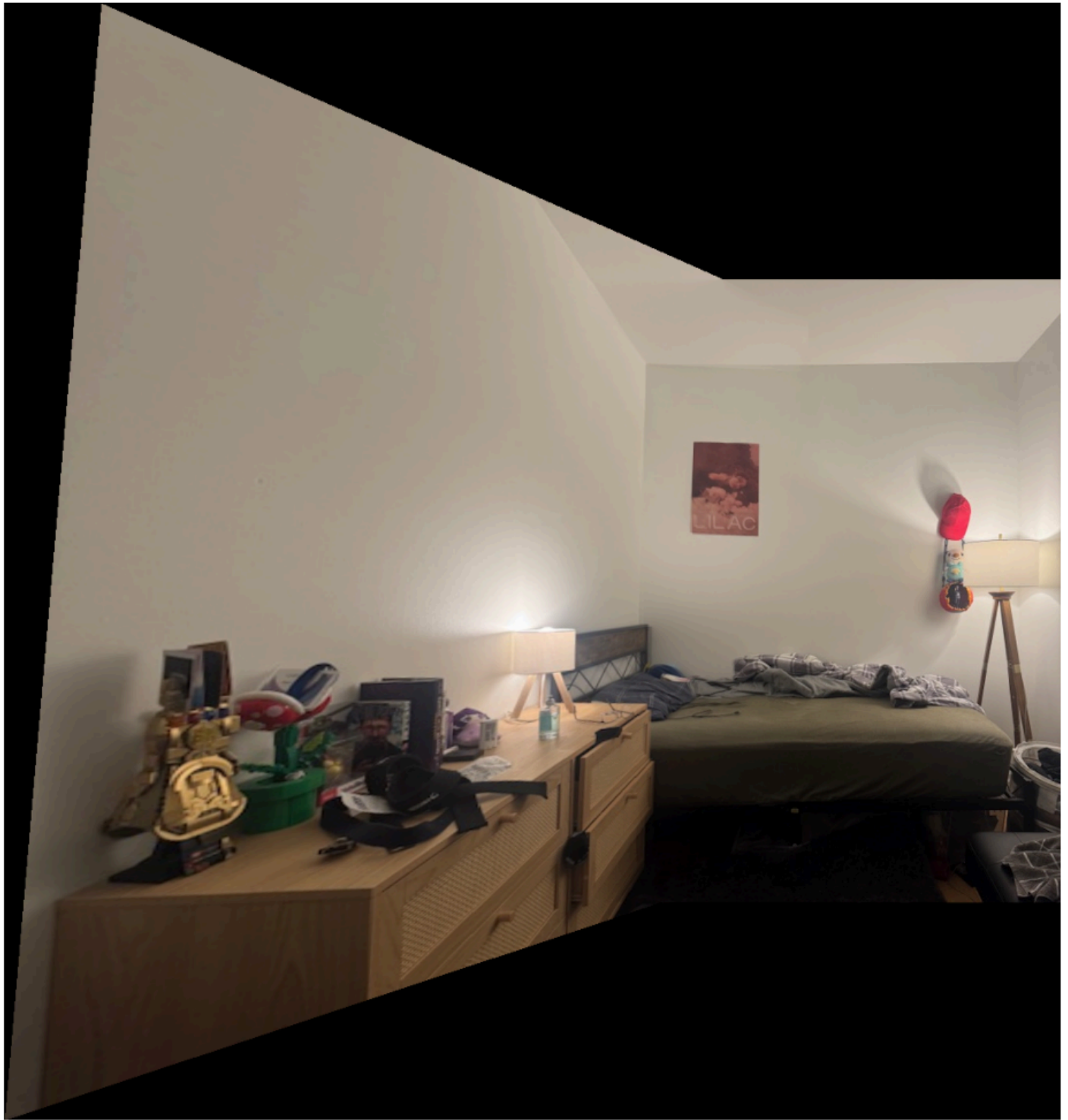




Room org



Room warp



Room Mosaic